# Selection, Production, and Properties of Regolith Polymer Composites for Lunar Construction

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*Abstract*— **NASA's Artemis Program seeks to establish a long-term presence on the Moon to enable scientific exploration and expand the cis-lunar economy by utilizing lunar resources through public-private and international partnerships. Over the next decades, a lunar spaceport will need to be established to provide the services and facilities that are necessary to achieve this goal. Robotic construction technologies using in-situ materials must be developed to build up enabling infrastructure such as launch/landing pads, blast protection, power/communications infrastructure, improved roads, improved operational surfaces, and radiation protection shelters. Kennedy Space Center's Granular Mechanics and Regolith Operations laboratory at Swamp Works has partnered with SpaceFactory and LERA Consulting Structural Engineers to develop the architectural and structural design of an unpressurized lunar shelter. The shelter, called Lunar Infrastructure Asset (LINA), is designed to support 2.3 m of regolith overburden to protect astronauts and surface assets from radiation, meteoroid impact, thermal gradients/cycling, and to withstand moonquakes. LINA's structural design criteria and the resulting structure design are detailed in separate papers. A Fused Filament Fabrication (FFF) construction process using regolith polymer composites has been developed and is detailed in a separate publication. This paper presents material formulations and selection rationale for each of the composite components that were examined. Formulations include 70:30, 80:20, and 85:15 wt.% mixture ratios of lunar mare simulant Black Point-1 (BP-1): Polylactic Acid (PLA), 80:20 Lunar Highlands Simulant-1 (LHS-1): PLA and an 80:20 BP-1: PLA formulation with a flow enhancer additive. Test samples were printed in simulated lunar dirty thermal vacuum conditions (-190 °C, 10-3 torr). A series of tests were performed on each formulation to characterize the achieved mixture ratio, mechanical strength properties, off-gassing products during vacuum printing, and porosity and density of printed products. The LHS-1: PLA formulation yielded an advantageous combination of properties and was used in a final test that additively constructed a sub-scaled LINA on regolith simulant in dirty vacuum conditions. The materials and construction process presented in this paper are considered to be at Technology Readiness Level (TRL) 5 with additional testing necessary to characterize long term lunar environmental exposure effects.**

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## 1. INTRODUCTION

NASA's Artemis Program seeks to establish a long-term presence on the Moon to enable scientific exploration and support the emergence of a cis-lunar economy utilizing lunar resources through public-private and international partnerships [1]. Over the next decades, a lunar spaceport will need to be established to provide the services and facilities that are necessary to achieve this goal. Robotic construction technologies using in-situ materials must be

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developed to build up enabling infrastructure such as launch/landing pads, blast protection, power/communications infrastructure, improved roads, improved operational surfaces and radiation protection shelters. Kennedy Space Center's (KSC) Granular Mechanics and Regolith Operations (GMRO) laboratory at Swamp Works has partnered with SpaceFactory and LERA Consulting Structural Engineers on a project called Relevant Environment Additive Construction Technology (REACT). REACT developed an architectural and structural design of an unpressurized protective shelter, Regolith-Polymer Composite (RPC) materials, and a Fused Filament Fabrication (FFF) construction process. The shelter, called Lunar Infrastructure Asset (LINA), is designed to support 2.3 m of regolith overburden and protect astronauts and surface assets from radiation, meteoroid impacts, thermal cycling/gradients, and withstand moonquakes.



**Figure 1. An artistic rendering of one concept of LINA's full scale construction process.**

Related publications present LINA's structural design criteria based on lunar environments, the resulting structural analysis and design using methodology based on terrestrial practices, and the additive construction equipment/processes used to build test samples and a sub-scale LINA in simulated lunar conditions [2] [3] [4].

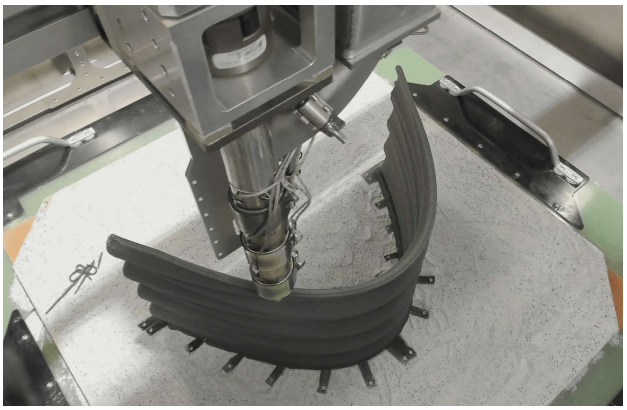


**Figure 2. A subscale LINA is additively constructed in simulated lunar conditions.**

This paper presents the regolith-polymer composite material formulations that were examined, the selection rationale for each component of the composite, RPC production methods and resulting characteristics of samples printed in both ambient and simulated lunar Dirty Thermal Vacuum (DTVAC) conditions.

## 2. Selection of Polymers for the Lunar Environment

A study was completed to compile and assess various polymers for their suitability for additive construction in lunar environments. The lunar environment is characterized by large swings in temperature (averages ranging from -212 to -91 °C at 85° latitude), hard vacuum ($10^{-12}$ to $10^{-14}$ torr) [5], galactic cosmic ray and solar particle event radiation, and meteorite impacts. These challenging environmental factors can subject polymers to thermal fatigue, mass loss due to outgassing, and UV degradation.

Considered polymers must be thermoplastics conducive to additive manufacturing and bond readily to the regolith filler. Challenges are presented by lack of convective cooling in vacuum, thermal extremes, high shrinkage rates of some polymers, and porosity resulting from processing in vacuum. Desirable properties include low outgassing rates, good mechanical properties throughout lunar temperature ranges, and UV resistance. A list of relevant material properties is provided below:

- Mechanical strength
- Elasticity
- Impact strength
- Coefficient of thermal expansion
- Thermal cycling fatigue life
- Molecular weight
- Glass transition temperature
- Melting temperature
- Specific heat capacity, enthalpy of fusion

A summary table of polymer material properties data and source data found during the study is provided in the Appendix.

Evaluated polymers were split into 3 categories as shown in Figure 3.

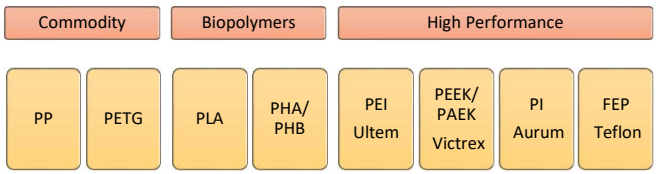



**Figure 3. Evaluated polymers by category.**

Commodity polymers consist of those which are commonly used in packaging and consumer applications. These polymers are inexpensive, readily available and may be recycled from packaging materials which may already be

present on spacecraft. However, these materials typically have average mechanical properties and are not intended for use in space environments. Evaluated commodity polymers include polypropylene (PP) and polyethylene terephthalate glycol (PETG).

Biopolymers are those polymers which can be sourced from renewable, biological processes. These polymers are relatively inexpensive (although 2-3x more expensive than commodity polymers), may be generated in-situ using plant growth or bioremediation processes, and are more sustainable than synthetic polymers. If produced in-situ, these polymers will require large quantities of water and otherwise habitable areas for growth and processing for infrastructure scale quantities. These polymers have limited data on their performance in vacuum and cryogenic environments, although the potential benefits merit further investigation in these areas. Evaluated biopolymers include Polylactic Acid (PLA), Polyhydroxyalkanoates (PHA) and Polyhydroxybutyrate (PHB).

High performance polymers are those which are specially formulated for space or other extreme environments and applications. These polymers may be certified and proven for space applications and may have known good performance in space environments. They typically have ~2-4x better mechanical properties than the other categories of polymers and may retain more toughness at cryogenic temperatures. In some cases, as in Fluorinated Ethylene Propylene (FEP), they may also exhibit significantly less outgassing in vacuum. High performance polymers may be ~4-50x more expensive than commodity and biopolymers. They typically require elevated processing temperatures (~400 °C) compared to the others (~200 °C). Evaluated high performance polymers include Polyetherimide (PEI), Polyetheretherketone (PEEK) / Polyaryletherketone (PAEK), Polyimide (PI), and Fluorinated Ethylene Propylene (FEP).

*Candidate Polymers*

Three polymers stood out for use in RPC. Polyetherimide, Polypropylene, and Polylactic Acid.

Polyetherimide (PEI), in particular Ultem 9085, is one of the few polymers approved by NASA for space applications. Ultem 9085 was selected as one of the materials for the MISSE-9 and MISSE-10 experiments, in which 3D printed samples were exposed to the low earth orbit space environment outside of the International Space Station to study the effects of radiation, thermal cycling and outgassing on material durability. Samples were exposed to the LEO environment for periods of 198 days (MISSE-9) to 250 days (MISSE-10). Results of these tests showed no significant decrease in tensile strength for these samples, although there was some reduction observed in failure strain [6].

Ultem 9085 was used to print the antenna array support structure for the six satellites in the COSMIC-2 weather satellite constellation. The structures were produced using a FFF method and replaced the standard antenna arrays which are machined from a silica fiber composite (astroquartz). Ultem 9085 underwent testing to meet NASA B/B1 flight requirements [7].

Polypropylene has common application in 3D printing and high melt flow index which facilitates high filler loadings. PP is commonly used in space mission packaging and waste has the potential to be repurposed as a construction/manufacturing material. Its low cost and recyclability make it appealing for terrestrial additive construction processes especially since it is hygroscopic which reduces processing steps and overall cost. PP has several drawbacks including a high CTE which tends to warp prints without a heated bed.

PLA has superior printability and relative low cost. PLA demonstrates superior cryogenic properties compared to PP. At 77 K PLA's ultimate tensile strength of 87 MPa is nearly twice PP's 45 MPa and is on par with Ultem, exhibiting 86% of Ultem's 101 MPa tensile strength (albeit with a much lower Young's modulus). PLA also retains a much greater percentage of its elongation to failure at cryogenic temperatures, dropping from 6.4% at room temperature to 5.7% at 77 K. A reduction of 10.9% versus PP's 96.4% reduction.

PLA suffers from several drawbacks which must be accounted for successful use. The first is that it has a tendency to absorb moisture, which may hydrolyze during extrusion and cause a degradation in mechanical properties. This drawback is mitigated in the vacuum environment of the Moon. PLA also typically has a low heat distortion temperature that would limit its performance in direct sunlight. Grades of PLA have been formulated to withstand temperatures of up to 134 °C [8]; further investigation is warranted into additives and other modifications to improve the heat resistance of PLA for lunar applications.

## 3. MATERIALS AND METHODS

Two rounds of materials development and testing were completed and are referred to as Alpha and Beta. The Alpha round was intended to establish basic feasibility. The Beta round was intended to characterize and improve a set of RPC formulations, and to select a final Beta material. The final Beta material properties were used to inform LINA's structural design which was constructed at subscale in simulated lunar conditions.

*Regolith Polymer Composite Feedstock Production*

All Regolith Polymer Composite (RPC) formulations were produced via a compounding process using a commercially available Brabender® corotating twin screw extruder and Plastic-Corder® Torque Rheometer drive shown in Figure 4. The extruder has a length/diameter (L/D) of 20 mm/ 40 D. The screw configuration included conveying, feeding, and kneading screw elements. The simulants (filler) and polymers (binder) were fed into the extruder using separate volumetric feeders. The weight ratio of filler to binder was set by

controlling the speed of each feeder. The extruder was normally operated at 36 RPM. The temperature of the extruder heat zones varied slightly across formulations and ranged from 190 to 210 °C from the feed zone to the die.




**Figure 4. Compounding equipment used to produce regolith-polymer composite.**

As the extruded filament leaves the die, it is cooled in a water trough, dried using compressed air jets, and then fed into a pelletizer. The take-up speed and blade rotation speed of the pelletizer are adjusted accordingly to place tension on the filament and cut it into pellets that are roughly 3 mm in diameter by 3 mm.

*Sample Construction Conditions*

The Advanced Regolith Ground Operations (ARGO) testbed was outfitted with an additive construction screw style pellet extruder to print test samples in ambient and simulated lunar DTVAC conditions as shown in Figure 5. A detailed description of the additive construction equipment and process is available in another publication [9] [4].

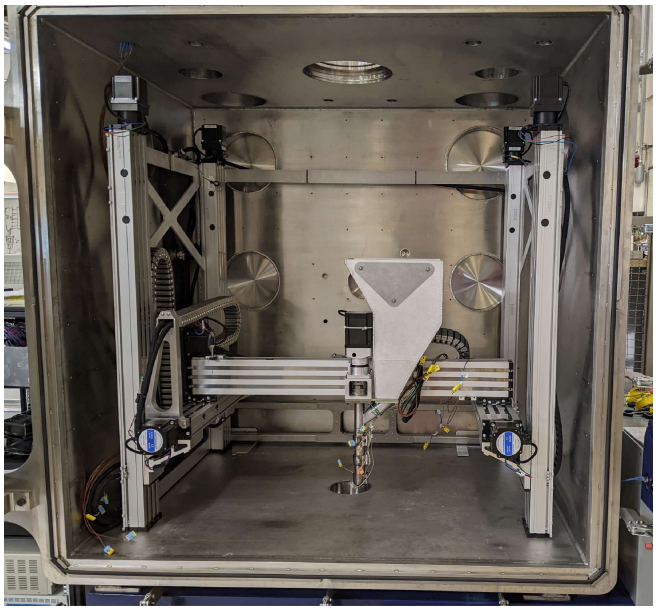


**Figure 5. The ARGO testbed outfitted with an additive construction extruder.**

*Alpha Materials*— For Alpha material samples printed in vacuum conditions, prints were done on a heated bed held at 100 °C. For BP-1: PP alpha material testing the bed was held at 100 °C and the extruder barrel at 175 °C. For BP-1: PLA alpha material testing the bed was held at 50 °C and the extruder barrel at 170 °C. Barrel and nozzle temperatures during steady state flow for both materials read approximately 5 °C less than the set temperature. Flow from the nozzle during construction was maintained at 60 $mm^3/s$. Vacuum was maintained at 0.250 torr.

*Beta Materials*— For Beta material samples printed in ambient conditions, prints were done on a heated bed held at 60 °C. The material was extruded with a barrel temperature of 175 °C for pure PLA and 190 °C for BP-1: PLA and LHS-1: PLA. The extrusion temperature was approximately 135 °C during steady state flow for PLA. Steady state temperatures for BP-1: PLA and LHS-1: PLA were 189 °C. Flow from the nozzle during construction was maintained at 60 $mm^3/s$.

Beta material samples were created under DTVAC conditions with a cryogenically cooled environmental shroud. This shroud was kept below -190 °C on all faces to simulate the loss of heat into space that would occur during construction on the lunar surface. A majority of the shroud was below -200 °C during operations, but areas furthest from the cryogenic cooling system reached -190 °C by the end of the operation due to radiative heating from the extruder and material test sample. The vacuum was maintained using a dual vane rotary roughing pump. The vacuum pressure started between $6x10^{-3}$ – $8x10^{-3}$ torr for each sample. During construction, the pressure rises in the chamber to 0.080 - 0.100 torr, likely due to sublimation of water vapor off the cryogenic shroud as the extruder and printed structure warms the shroud. Figure 6 shows this increase in pressure. The pressure increases to ~0.8 torr and then begins to asymptotically stabilize, likely due to a lack of water vapor remaining on the cryogenic shroud to sublimate. The material was extruded through a screw style extrusion nozzle with a barrel temperature of 200 °C and the media extrusion temperature approximately 189 °C during steady state flow. Flow from the nozzle during construction was maintained at 60 $mm^3/s$.

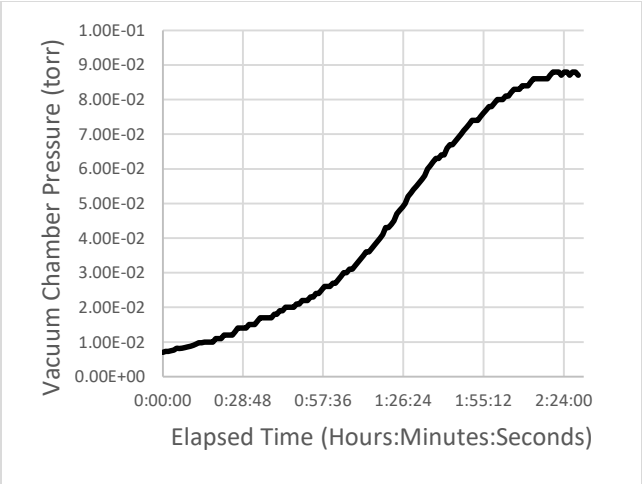



**Figure 6. Vacuum chamber pressure during sample construction.**

To produce Beta material samples for strength testing, a simple square tube geometry with open ends on the top and bottom was printed. This shape, shown in Figure 7, was chosen to produce the greatest total uninterrupted flat side surface areas that would be exposed to the cold shroud. This tube was produced using one continuous extrusion of material to avoid discontinuities which is a method of additive construction known as spiral outer contour or vase style.

Five square tubes were printed using the following formulations for Beta materials:

- BP-1: PLA 70:30 wt.%
- BP-1: PLA 80:20 wt.%
- BP-1: PLA: CAI Performance, CP-L01 Flow Improver 80:18.5:1.5 wt.%
- Lunar Highlands Simulant-1 (LHS-1): PLA 80:20 wt.%
- BP-1: PLA 85:15 wt.%

During sample construction, the build surface was measured to be -90 °C to -100 °C due to radiative and conductive losses to the cryogenic shroud. The square tube structures were unable to bond to the PLA build surface and instead were mechanically fastened by extruding the initial tube layer on and around hook shaped nylon barbs that were mechanically installed onto the build plate. This created a firm bond without inducing warping or damage to the tube beyond the first layer. Additional details on system parameters, environment conditions, and thermal vacuum hardware are available in a separate publication [9].

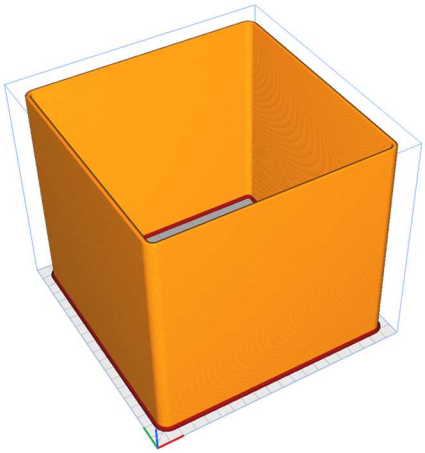


**Figure 7. Slicer generated preview of the material testing sample square tube geometry.**

Each side of the tube was 184 mm wide and 170 mm tall with a nominal tube wall thickness of 4 mm. The corners of the tube were rounded with an 8 mm radius. Figure 8 shows the five formulation TVAC printed cubes. Each sample was separated into four flat panels by cutting along each side of the radiused corners. The corner sections and pellets were reserved for analysis of achieved mixture ratio, density, porosity, morphology, and hydrogen bonding.




**Figure 8. TVAC printed test cubes prior to sectioning.**

*Beta Materials Sample Sectioning*

The flat panels cut to size for flexural, compression, and tension tests. This was accomplished utilizing a tile saw with a diamond coated blade cooled with water chilled to 10 °C to prevent bonding of the cut surface. Four samples of each material were tested in each of layer orientations per test.

*Beta Materials Mechanical Testing*

Early structural designs of LINA were controlled by flexural strength and modulus of elasticity, therefore these properties

were tested by SpaceFactory on each of the Beta formulations to inform the selection of the final Beta material. Afterwards, SpaceFactory performed compression and tensile testing on the remainder of the formulations. The formulation selected as the final Beta material was sent to an independent test laboratory to establish its mechanical properties more formally. Testing by SpaceFactory was done as follows.

Flexural tests were conducted according to ASTM D790 Standard Test Methods for Flexural Properties. Flexural test specimens were cut to rectangles of approximately 18 x 100 mm, approximately 6 mm in thickness (the print bead width). A three-point bend test fixture with a span of 80 mm was used. Specimens were tested in two different orientations: force parallel to layer lines (0 degrees), and perpendicular to layer lines (90 degrees). Three to four samples of each formulation in each orientation were tested. Force data was acquired utilizing a Mark-10 ESM1500S with a maximum force of 10 kN and accuracy of 0.1%.

Compression tests were conducted according to ASTM D6641 Standard Test Method for Compressive Properties of Polymer Matrix Composite Materials Using a Combined Loading Compression (CLC) Test Fixture. Compression test specimens were cut to squares of approximately 18 mm x 18 mm, approximately 6 mm in thickness (the print bead width). These sample sizes, though much smaller and of different shape than standard ASTM tests, were chosen because their maximum compressive load remained at approximately 75% of the maximum force provided by the Mark-10 ESM1500S. Specimens were tested in two different orientations: compression force parallel to layer lines (0 degrees), and perpendicular to layer lines (90 degrees). Three to four samples of each formulation in each orientation were tested.

Tensile testing was conducted according to ASTM D3039 Standard Test Method for Tensile Properties of Polymer Matrix Composite Materials. Tensile test specimens were cut to rectangles of approximately 20 x 50 mm approximately 6 mm in thickness (the print bead width). These samples were then notched on both sides with a 2.5 mm radius utilizing a round metal file to create the gauge region. The samples were then pulled with a combination of tensile gripping calipers and a standard vice. Tests were conducted in two different orientations: tension force parallel to layer lines (0 degrees) and perpendicular to layer lines (90 degrees). Four samples of each formulation in each orientation were tested. Strain data pulled for these tests was highly varied because of difficultly with available tools due to the very small amount of deflection experienced by the material and an inability to create a perfect and reproducible yielding zone.

The independent test laboratory completed mechanical testing on the final Beta material according to Table 1. All testing was performed at prevailing laboratory conditions (temperature, humidity) in accordance with applicable ASTM specifications. At least three samples of each formulation in each orientation were tested. Failure modes were observed and documented.

**Table 1. Independent laboratory mechanical tests.**

| Test Description | Test Specification | Layer Orientation | Samples per Panel |
|---|---|---|---|
| Tensile Properties | ASTM D3039 | 0º, 90º | 3, 3 |
| Compression Properties | ASTM D6641 | 0º, 90º | 3, 3 |
| Tensile Properties | ASTM D5379 | 0º, 90º | 3, 3 |
| Flexural Properties | ASTM D790 | 0º, 90º | 3, 3 |

*Beta Materials Chemistry*

Thermogravimetric Analysis (TGA) using a TA TGA5500 was performed on all materials to confirm chemistry and mixture ratios. Approximately 10 µg was used for each sample.

Ash testing was performed as an alternative method to confirm mixture ratios in all pellet and print samples. 1 to 3 grams of each sample was weighed and added to a crucible of known weight. The samples are then placed in an oven at 600ºC for 1 hour. Once cool, the remaining material in each crucible is weighed and recorded. The remaining material is assumed to be filler, so the material composition can be determined by comparing the starting sample weight to the remaining sample weight.

Fourier-transform infrared (FT-IR) spectra of PLA and 80:20 LHS-1: PLA pellets were collected using a Nicolet iS50 FT-IR equipped with a Nicolet Continuum™ FT-IR Microscope.

Residual Gas Analysis (RGA) was performed using an INIFICON Transpector® MPH Gas Analysis System to identify and quantify residual gases, vapors, or contaminants that may be released during printing under lunar-like thermal vacuum conditions. A vacuum interface was installed on the vacuum chamber and the RGA was connected both electrically and through the interface allowing it to sample the chamber gases prior to printing for baseline conditions and during vacuum printing. The setup is shown in Figure 9.

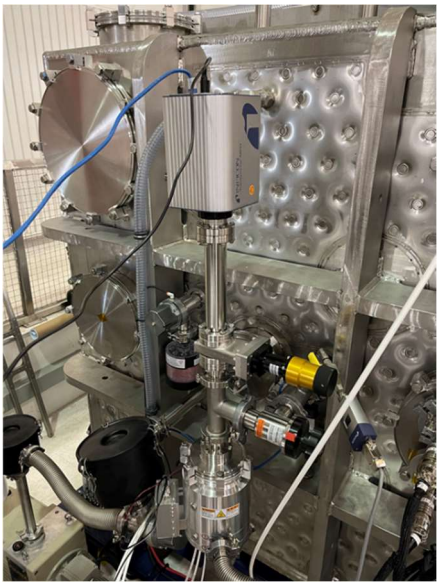


**Figure 9. Residual Gas Analyzer (RGA) and vacuum chamber setup**

*Beta Materials Density and Porosity*

The density of the formulations was measured using both a standard volumetric displacement method and calculated from volume data using an Ultrapyc 1200e pycnometer.

An UltraPyc 1200e gas pycnometer designed by Quantachrome Instruments was used to measure the volume and density of the solid materials. Nitrogen was chosen as the inert gas for its ability to permeate small pores within the material.

Density was also measured using a traditional volumetric displacement method. Pellet samples were weighed and added to a graduated cylinder filled with a known volume of 2-propanol to measure their total volume via the change in volume.

CT-Scans were completed on samples cut from each of the five square tubes printed in thermal vacuum and square tubes of BP-1: PLA 80:20 wt.%, LHS-1: PLA 80:20 wt%, and 100% PLA printed in ambient conditions. The system used was the North Star Imaging, model X5000. It is a 225KV system with a Perkin Elmer 100-micron detector panel. We utilized three main programs. efX-DR (to acquire the digital radiographs) and then efX-CT (to reconstruct and manipulate the computed tomography volume). The program used to quantify the porosity was Volume Graphics, Studio Max, with the Porosity and Inclusion add-on module. These programs provide a 3D reconstruction of the scanned sample with values corresponding to relative density for each voxel. Filtering of these relative density values is used to determine which parts of the material is solid which parts are voids.

Porosity calculations of the materials from data obtained using pycnometer and displacement data were done by comparing the theoretical density to the measured density. For calculations, the TGA measured mass fractions of PLA and regolith in the formulation were converted to volume percent using the below equation:

$$\boldsymbol{V_A = \frac{W_A \rho_A}{W_A \rho_A + W_B \rho_B}} \quad (1.1)$$

Where W is the weight percent of component A or B and ρ is the density of component A or B.

The theoretical densities were calculated with the rule of mixtures below:

$$\boldsymbol{\rho_{theor.} = \frac{\rho_{PLA} V_{PLA} + \rho_{Regolith} V_{Regollth}}{100}} \quad (1.2)$$

Where $\rho_{PLA}$ is the density of raw PLA pellets, $V_{PLA}$ is the volume percent of PLA in the formulation, $\rho_{Regolith}$ is the density of pure regolith, and $V_{Regolith}$ is the volume percent of regolith in the formulation.

The porosity was then calculated with this theoretical density and the measured density:

$$Porosity\ (\%) = 1 - \left(\frac{\rho_{meas.}}{\rho_{theor.}}\right) * 100\% \quad (1.3)$$

Where $\rho_{meas.}$ is the density measured from either the displacement or pycnometer methods.

Porosity from the data obtained from CT-Scnas was calculated using equation below:

$$Porosity\ \% = \frac{v_{pores}}{v_{pores} + v_{material}} * 100\% \quad (1.4)$$

Where $v_{pores}$ is the volume of pores and $v_{material}$ is the volume of material.

*Beta Materials Microscopy*

Scanning Electron Microscopy (SEM) was completed on the fracture surfaces of tensile test specimens. A Jeol-I1800 Field Emission SEM was used. Samples were mounted on stubs with carbon tape and sputter coated with gold using a Denton Vacuum Desk IV prior to imaging.

## 4. Test Results and Discussion

As stated earlier, two rounds of materials development were completed and are referred to as Alpha and Beta. The intent of the Alpha material development was to demonstrate basic feasibility of RPC production in lab conditions, printing processes in rough vacuum, and to establish baseline material properties. The Beta materials development round improved RPC by evaluating the effects of higher regolith wt.% loading levels, alternative simulants, a processing aid additive, and lower temperatures and pressures during the print process.

Initially, the Beta materials underwent mixture ratio characterization along with flexural strength and modulus of elasticity testing to inform the selection of the final Beta material within schedule constraints. Afterwards, SpaceFactory performed compression and tensile testing on

all the formulations. The formulation selected as the final Beta material was sent to an independent test laboratory to determine strengths and modulus of elasticity for compression, tension, flexure, and shear loads. Subsequent tests characterized density, porosity, morphology, vacuum print off-gas products and hydrogen bonding.

*Alpha Materials Development*

Polyetherimide (PEI) is an obvious choice as the binder for RPC, however the project was unable to use it because of extrusion equipment temperature limitations and cost.

Polypropylene (PP) (COPYLENE CH550LN) was the initial choice of binder for Alpha materials [10]. A Polypropylene-Graft-Maleic Anhydride (PP-MAH) compatibilizer (SCONA TPPP 9212 GA) was used to promote adhesion between the polymer matrix and regolith filler [11].

Black Point-1 (BP-1) was selected as the regolith simulant filler for the Alpha materials because it is a suitable general-purpose simulant with comparable particle size distribution and geotechnical properties to lunar regolith [12]. Additionally, it is a low-cost simulant that is abundantly available at the GMRO laboratory.

Alpha material production and testing initially focused on a 50:48:2 BP-1: PP: PP-MAH wt.% ratio formulation. A 50% loading level of filler was selected for basic feasibility testing of material production and vacuum chamber extrusion since these processes can be significantly more challenging with higher filler loading levels. BP-1 was dry sieved to pass a 500 µm screen. The PP-MAH additive was supplied in PP pellet form with an MAH content of 1.8%. The additive was pre-mixed into the pure PP pellets before adding to the feeder.

Additive construction tests at 0.30 to 0.20 torr exhibited a highly porous and brittle composite as can be seen in Figure 10 (a). Several rounds of testing with various drying procedures, print settings and additive content were examined to determine the source of the porosity. PP was ruled out as the source due to a pore free vacuum printing test using undried neat PP as shown in Figure 11 (left). A test mixture of 50:50 PP: PP-MAH exhibited significant porosity far exceeding other compositions as seen in Figure 11 (right) indicating that the additive was a source of porosity. However, a 50: 50 PP: BP-1 wt.% compositions that did not contain the additive still had significant voids in all tests. It is thought that porosity could also be attributed to residual moisture in the BP-1 despite drying it in a convection air oven at 204 °C for at least 2 hours before compounding and drying the composite in vacuum at ambient temperature for at least 12 hours before printing. It is possible that improved simulant drying procedures could alleviate porosity in PP based formulations, however further assessment was not pursued because PLA formulations were found to have superior performance (Figure 10 (b)) as discussed below.

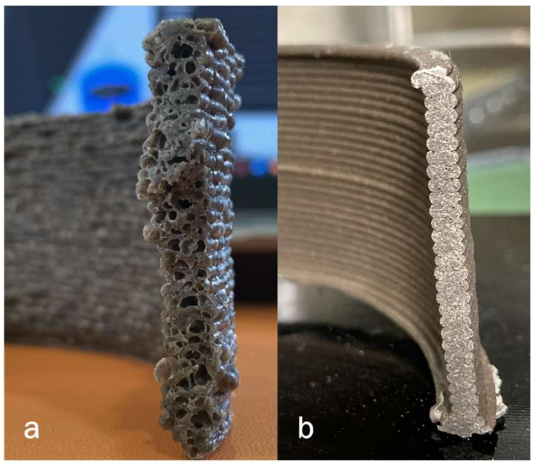



**Figure 10. Significant porosity is observed BP-1: PP (a), but not in BP-1: PLA (b).**



**Figure 11. A neat PP print has minimum porosity (left). A PP-MAH test print exhibits significant porosity (right).**

Polylactic Acid (PLA) (Luminy LX175) was selected as the binder in a second version of the Alpha materials. PLA was compounded with BP-1 at 50:50 wt.%. The BP-1 simulant was dried as described above and the PLA was dried in a PrintDry filament drier at 60 °C for at least 4 hours. The composite was dried in the filament drier for at least 3 hours at 60 °C prior to additive construction tests. Using PLA in vacuum printing at ~$10^{-3}$ torr significantly mitigated porosity as seen in Figure 10 (b). It is not clear why PLA based formulations exhibited significantly less porosity that PP formulations even though similar drying procedures were used. PLA has greater gas permeability that PP, which could allow moisture to escape and prevent pore formation. These

tests proved basic feasibility of the PLA based materials and construction process.

*Beta Materials Drying Procedure*

All PLA was dried for a minimum of 4 hours at 60 °C in a filament drier and all simulants were dried for at least 2 hours at 204 °C in a convection oven before compounding. All PLA:BP-1 and PLA:LHS-1 formulations were dried prior to printing using a filament drier for a minimum of 4 hours at 60 °C.

*Initial Characterization for Beta Material Selection*

*Target Versus Achieved Mixture Ratios*— Pellet samples of each formulation were examined to determine the achieved mixture ratio by TGA and ash testing. Table 2 provides a summary of the achieved mixture ratios for each formulation and an error percentage based on the average of the TGA and ash test values compared to the target values. TGA and ash test values show general agreement. The naming convention for materials throughout the paper uses the target wt.% values rather than achieved for simplicity. The relatively high error in target vs. achieved mixture ratios indicates that feeder system settings and/or methods must be improved to provide a more accurate mixture ratio.

*Ultimate Flexural Strength*— Initial tests showed that 80:20 BP-1: PLA demonstrated the highest ultimate flexural strength value for the test samples that were cut perpendicular to the grain (17.8 MPa). 80:20 LHS-1: PLA had the highest ultimate flexural strength value for the test pieces that were cut parallel to the grain (33.4 MPa); however, 80:20 BP1:PLA was 2nd only by 1.03 MPa or roughly 3%.

*Modulus of Elasticity*— Initial tests showed that 80:20 LHS-1: PLA displayed the highest flexural modulus of elasticity when loaded both perpendicular (3.68 GPa) and parallel (7.43 GPa) to the grain.

*Strength Properties*— Though the 80:20 BP-1: PLA and 80:20 LHS-1: PLA formulations exhibited the best (and comparable) overall flexural strength properties, the evolving analysis of the thin shell LINA arch structure revealed that internal stresses were relatively low compared to the material strength values gathered through testing. The LINA design is driven more by material modulus rather than strength to reduce total structural movement during moonquakes and overburden load [3]. Since, the 80:20 LHS-1: PLA formulation showed the greatest flexural modulus, it became the leading candidate for selection as the final Beta material.

**Table 2. Target versus achieved mass percent (wt.%) mixture ratios.**

| Material | Target wt.% Ratio | Actual wt.% Ratio, TGA | Actual wt.% Ratio, Ash | Averaged % Deviation from Target |
|---|---|---|---|---|
| PLA pellet | 0:100 | - | 0:100 | 0.0 |
| CP-L01 | 0:100 | - | 0:100 | 0.0 |
| BP-1 | 100:0 | - | 99:1 | 1.3 |
| LHS-1 | 100:0 | - | 100:0 | 0.0 |
| BP-1: PLA pellet | 70:30 | 61:39 | 62:38 | 12.3 |
| BP-1: PLA pellet | 80:20 | 70:30 | - | 12.5 |
| BP-1: PLA: CP-L01 pellet | 80:18.5:1.5 | 72:28* | 71:29* | 8.7 |
| LHS-1: PLA pellet | 80:20 | 75:25 | 74:26 | 7.0 |
| BP-1: PLA pellet | 85:15 | 78:22 | 78:22 | 8.5 |
| BP-1: PLA TVAC | 70:30 | - | 62:38 | 11.8 |
| BP-1: PLA TVAC | 80:20 | - | 71:29 | 11.9 |
| BP-1: PLA: CP-L01 TVAC | 80:18.5:1.5 | - | 71:29* | 9.7 |
| BP-1: PLA TVAC | 85:15 | - | 77:23 | 9.2 |
| BP-1: PLA Ambient | 80:20 | - | 70:30 | 12.1 |
| LHS-1: PLA Ambient | 80:20 | - | 75:25 | 5.7 |
| PLA Ambient | 0:100 | - | 0:100 | 0.0 |

* CP-L01 is consumed during the compounding process.

*Comparative Analysis for Beta Material Selection*

*Simulant Selection*— The project selected the proposed Artemis lunar south pole connecting ridge location as the proposed site for LINA. Therefore, a simulant's suitability for representing lunar highlands regolith was the primary factor for simulant selection. Lunar Highlands Simulant-1 (LHS-1) is a high-fidelity mineral-based highlands simulant that is composed primarily of anorthosite (74.4 wt.%) and glass-rich basalt (24.7 wt.%). Its particle size distribution and mineralogy are tailored to approximate that of lunar regolith in highlands and polar regions [13]. Black Point-1 (BP-1) is a basalt-based mare simulant that captures the size distribution and geotechnical properties of lunar regolith in Mare regions [12].

LHS-1 provides a better representation of lunar highlands regolith than BP-1 because of its higher plagioclase (anorthosite) content. Anorthosite is harder than basalt which can affect processing parameters like energy consumption, friction, and wear.

It should be noted that neither simulant includes agglutinates, nanophase iron, or solar wind deposited elements like actual lunar regolith, and both have glass content that is somewhat lower than most lunar regolith samples. Higher fidelity simulants were not within the cost constraints of the project.

*Selected Beta Material*— 80:20 LHS-1: PLA was chosen as the Beta material due to its superior flexural modulus and greater accuracy as a lunar highlands regolith simulant.

*Additional Strength Properties Testing of Beta Materials*

*SpaceFactory Testing*— All the formulations underwent further strength properties testing at SpaceFactory's facilities with the results presented in

Table 3. A comparison of the flexural and compressive modulus of elasticity data is provided in Figure 12. A comparison of the flexural, compressive and tensile strength properties is provided in Figure 13.

*External Lab Testing*— The independent lab mechanical testing included tensile, compression, flexural, and shear. A summary of test results for loading in 0º and 90º orientations is provided below in Table 4. Figure 14 and Figure 15 show a comparison of the external lab testing strength and moduli data, respectfully. An image of the post-test flexural test samples is provided in Figure 16.

**Table 3. Summary of the material testing results performed at SpaceFactory**

| Composite | Orientation | Avg Ultimate Flexural Strength (MPa) | Avg Flexural Modulus (GPa) | Avg Ultimate Compressive Strength (MPa) | Avg Compressive Modulus (Gpa) | Avg Ultimate Tensile Strength (MPa) |
|---|---|---|---|---|---|---|
| 70/30 BP-1/PLA | 0º | 30.17 | 3.22 | 43.83 | 3.28 | 10.92 |
| | 90º | 17.50 | 1.93 | 48.85 | 5.38 | 14.19 |
| 80/20 BP1/PLA | 0º | 32.00 | 4.08 | 46.05 | 4.43 | 10.85 |
| | 90º | 17.70 | 2.87 | 47.88 | 8.95 | 14.63 |
| 80/20 BP1/PLA w/ Additive | 0º | 32.03 | 4.68 | 45.08 | 4.04 | 11.70 |
| | 90º | 18.20 | 3.13 | 46.80 | 7.73 | 15.36 |
| 85/15 BP1/PLA | 0º | 22.58 | 5.71 | 43.40 | 6.30 | 5.18 |
| | 90º | 12.53 | 2.63 | 44.85 | 8.05 | 11.40 |
| 80/20 LHS-1/PLA | 0º | 34.40 | 7.43 | - | - | - |
| | 90º | 14.90 | 3.68 | - | - | - |

**Table 4. Summary of the Selected Beta Material (80:20 LHS-1: PLA) Properties as tested by an independent laboratory.**

| Property | 0º | | 90º | |
|---|---|---|---|---|
| | Average | Std Dev. | Average | Std Dev. |
| Tensile Strength (MPa) | 15.2 | 0.3 | 9.3 | 0.3 |
| Tensile Modulus (GPa) | 7.0 | 0.4 | 5 | 1 |
| Compression Strength (MPa) | 50 | 2 | 46.3 | 0.2 |
| Compression Modulus (GPa) | 7.5 | 0.4 | 6.41 | 0.07 |
| In-plane Shear Strength (MPa) | 17 | 2 | 14 | 1 |
| In-plane Shear Modulus (GPa) | 2.07 | 0.07 | 1.9 | 0.3 |
| Flexural Strength (MPa) | 24 | 3 | 16.6 | 0.8 |
| Flexural Modulus (GPa) | 5.3 | 0.2 | 3.7 | 0.6 |

*Strength Properties Discussion*— In general, the 3D printed samples under examination displayed anisotropic characteristics, meaning the test results for 0º and 90º samples varied due to the differing alignment of the printed grain relative to the applied force. The samples showed enhanced strength when the force was applied parallel to the grain, highlighting the anticipated directional dependence of their mechanical properties.

Compared to SpaceFactory's testing, the independent lab showed a lower flexural strength in the 0º and 90º orientations at 16.55 and 23.86 MPa respectfully. The flexural modulus in the in 0º orientation was also reduced to 5.31 GPa, however, in the 90º orientation it remained essentially the same at 3.65 MPa. The independent lab's compressive strength properties were the highest recorded in the 0º orientation at 49.22 MPa and comparable to SpaceFactory's values in the 90º orientation at 46.33 MPa. Compressive modulus was comparable in both 0º and 90º at 7.45 and 6.41 MPa respectfully.

For comparison, readily available lumber used in typical wood construction generally exhibits a modulus of elasticity of 6-10 GPa and a flexural strength of 4-8 MPa. Typical flexural strength of unreinforced concrete is similar and are in the range of 3-5 GPa. In comparison, the 80:20 LHS-1: PLA test specimens demonstrated comparable modulus to lumber, especially those that were cut parallel to the grain, and demonstrated significantly higher flexural strengths, in comparison to both lumber and concrete, that is on the order of one-tenth that of steel. These findings highlight the potential suitability and advantages of the material for lunar construction. However, the material properties reported here result from testing in ambient environments. An analysis of the expected temperatures of the materials on the Moon is necessary along with testing across the relevant thermal range.

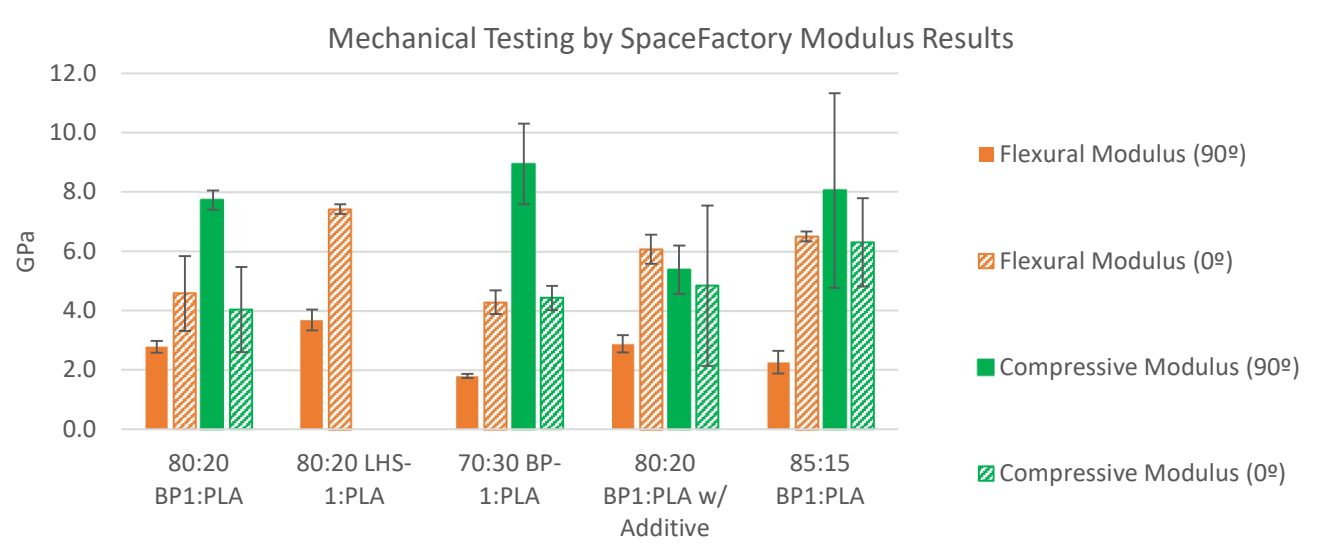


**Figure 12. Comparison of flexural and compressive modulus of elasticity data collected by SpaceFactory.**

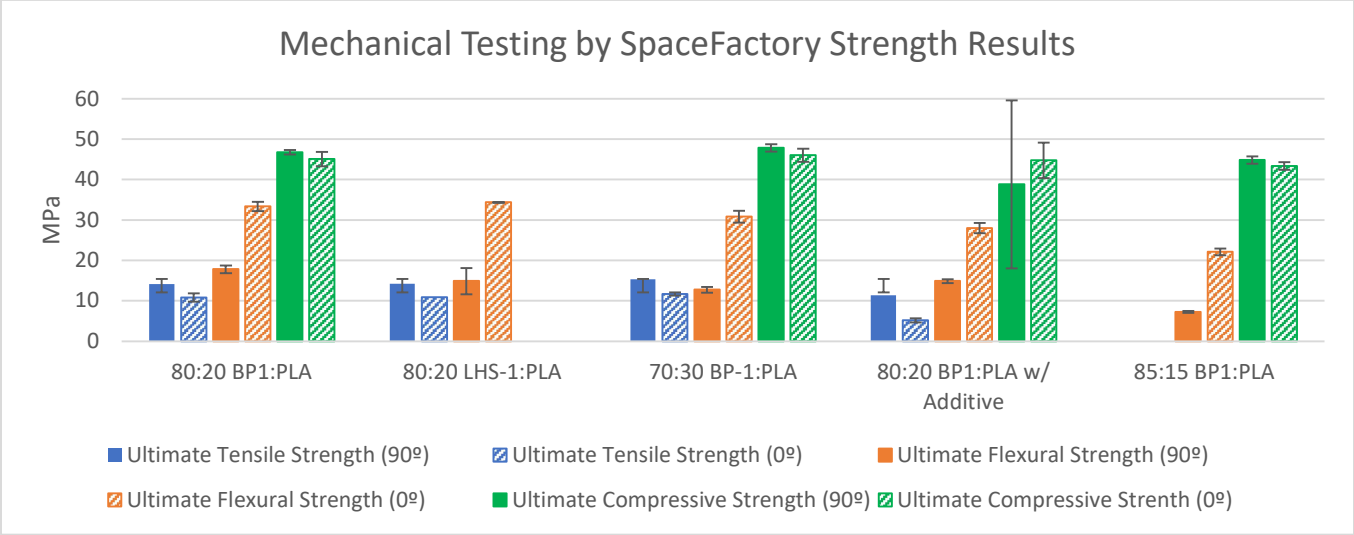



**Figure 13. Comparison of flexural, compressive, and tensile strength data collected by SpaceFactory**.

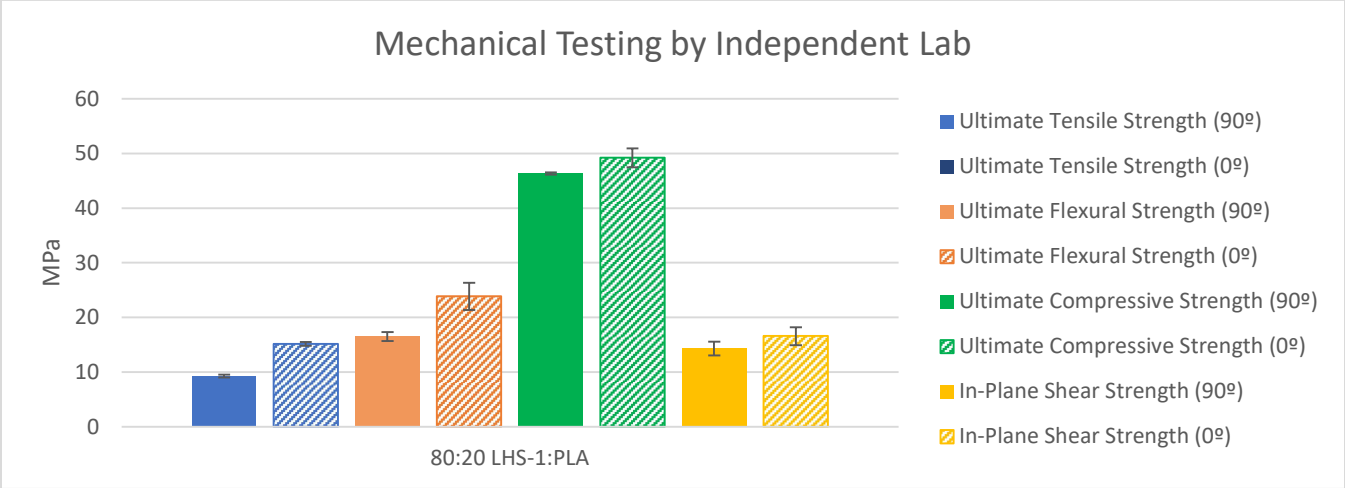



**Figure 14. Comparison of strength measurements collected by the independent laboratory.**

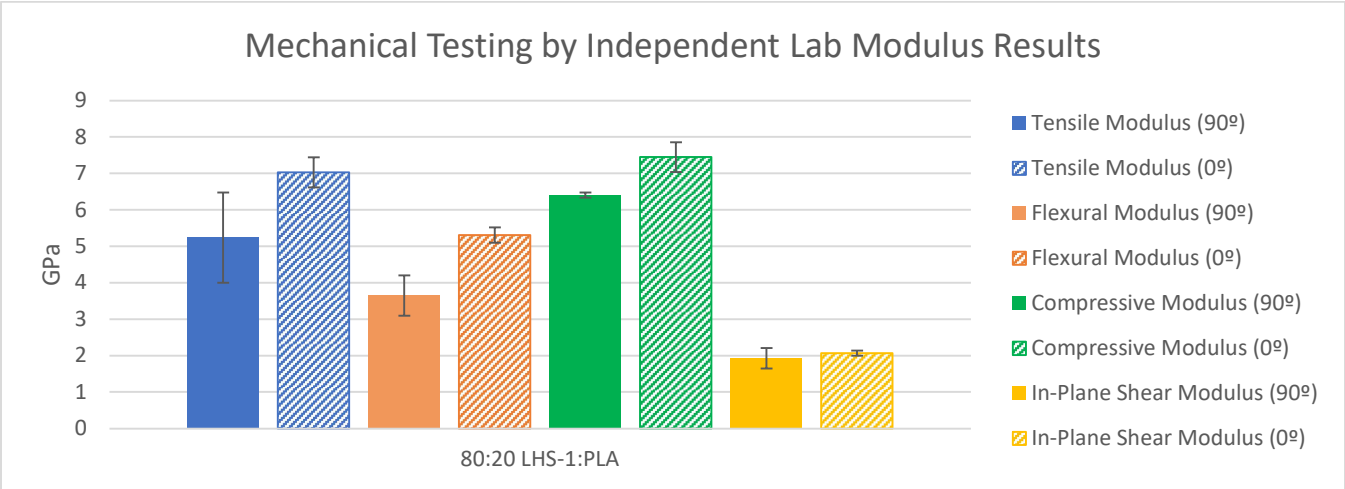



**Figure 15. Comparison of modulus measurements collected by an independent laboratory.**

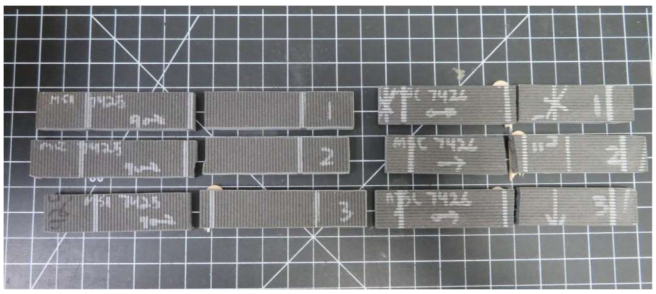


**Figure 16. Failure modes of flexural test specimens.**

Additionally, the properties of these 3D printed materials are not solely reliant on their intrinsic material properties but are also significantly influenced by various factors involved in the printing process. Parameters such as processing temperatures, print speeds, and extrusion rates play pivotal roles, as well potential induced defects, notably pore formation may occur due to off-gassing during the print. Because of these factors the material properties will depend on the construction system characteristics and process settings.

*Density Characterization of Beta Materials*

A summary of the densities of individual components of the composites are provided in Table 5. The BP-1 particle density measured by pycnometer is assumed to be in error because it is significantly greater than the density range of basalt which is 2.8- 3.0 g/cm$^3$. Table 6 provides density data for each of the formulations measured by pycnometer and displacement methods in pellet, ambient print, and TVAC print forms. Again, some of the pycnometer values are assumed to be in error because the composite densities approach, and exceed, simulant densities which is not possible at the measured mixture ratios. In general, density from the volumetric displacement method likely yields a more accurate composite density because the pycnometer gathers its sample volume data using a gas which can penetrate more of the voids on the interior of the pellets while the volumetric displacement method uses a liquid which cannot easily fill the interior pellet voids. Additionally, polymers are typically prone to absorption of gas. In both above cases, the volume filled by pycnometer gas will be higher than the actual sample volume resulting a higher sample density compared to the displacement method and reality. Furthermore, gas pycnometer measurements can be sensitive to factors including fill fraction of the chamber, temperature, and gas used. We present the pycnometer data for completeness, but do not use it in further analysis (except in noted cases). The selected Beta formulation (80:20 LHS-1: PLA) density was reported as 1.80 g/cc for use in LINA structural design and analysis tasks of the project in [3].

**Table 5. Individual composite component densities measured by pycnometer and displacement methods.**

| Material | Pycnometer Density (g/cc) | Displacement Density (g/cc) |
|---|---|---|
| Pure PLA | 1.25 | 1.28 |
| Pure BP-1 | 3.87* | 2.75 |
| Pure LHS-1 | 2.55 | 2.64 |
| CP-L01 Modifier | 1.87 | 1.22 |

*Assumed measurement error.

A comparison of the displacement densities of each formulation when in pellet, printed in ambient, and printed in TVAC is presented in Figure 17. The data reveals that the density of the formulations was the same or reduced following printing. This is expected because as the material is extruded by the print head, the polymer goes through additional consolidation from the forces exerted on it. This densification is increased by the subsequent layer being printed and reheating and compressing the previous layers. Inherent variability in any manufacturing process, including printing, should be considered. Even minor variations in the actual printing process, such as slight changes in temperature or print speed, could lead to density discrepancies.

The consistency of the trend across different mixture ratios implies that the printing of dense material in thermal vacuum is achievable. The remaining porosity in the material following printing can likely be furthered reduced with more optimization to pellet extrusion and printing parameters. Further discussion of porosity is in the following section.

*Porosity Characterization of Beta Materials*

Collected displacement data of pellets and printed parts was used along with mass fractions from TGA to calculate porosity and understand how void content changes across simulants, mixture ratios and between processing steps. The estimated porosity values resulting from these methods are shown in Table 7.

**Table 6. Density of PLA and each formulation in pellet, ambient print, and TVAC print form measured by pycnometer and displacement methods.**

| Composite | Pellet Densities (g/cc) | | | Ambient Print Densities (g/cc) | | TVAC Print Densities (g/cc) | | |
|---|---|---|---|---|---|---|---|---|
| | Pyc. | Pyc. Std. dev. | Disp. | Pyc. | Disp. | Pyc. | Pyc. Std. Dev. | Disp. |
| Pure PLA | 1.3 | 0.1 | 1.28 | - | 1.26 | - | - | - |
| 70:30 BP-1: PLA | 2.16 | 0.07 | 1.45 | - | - | 1.7 | 0.1 | 1.44 |
| 80:20 BP-1: PLA | 2.31* | 0.02 | 1.56* | - | 1.82 | 1.92 | 0.08 | 1.79 |
| 80:18.5:1.5 BP-1: PLA: CP-L01 | 2.53* | 0.04 | 1.73 | - | - | 1.91 | 0.04 | 1.78 |
| 80:20 LHS-1: PLA | 1.96 | 0.05 | 1.80 | - | 1.82 | 2.0 | 0.2 | - |
| 85:15 BP-1: PLA | 3.2* | 0.1 | 1.93 | - | - | 2.0 | 0.2 | 1.93 |

*Assumed measurement error.

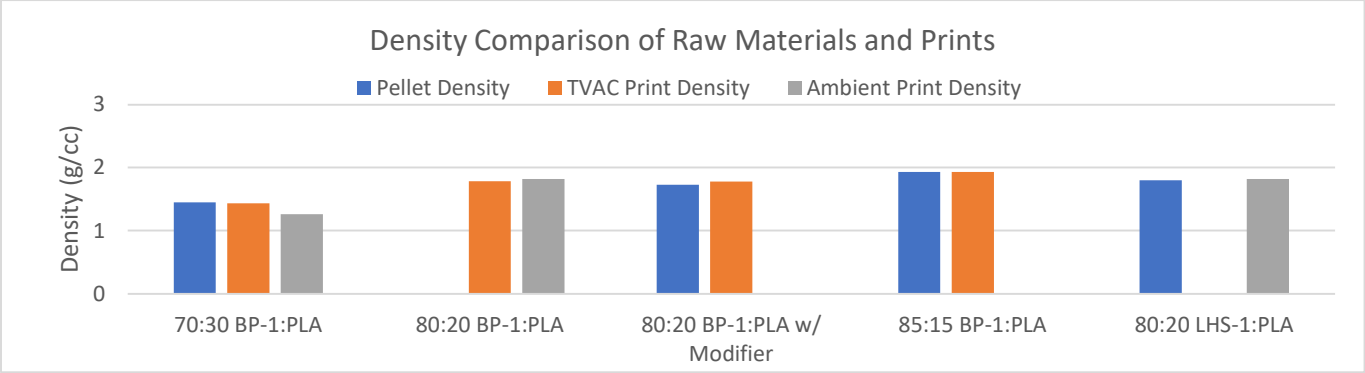



**Figure 17. Displacement density of pellets, ambient prints and TVAC prints showing that the environmental conditions do not have a significant effect.**

**Table 7. Porosity calculated from displacement density of the formulations in the form of pellets, ambient prints, and TVAC prints from calculated and CT-Scan data.**

| Composition Type | Pellet Porosity (%) | Ambient Print Porosity (%) | TVAC Print Porosity (%) | Ambient Print CT-Scan Porosity (%) | TVAC Print CT-Scan Porosity (%) |
|---|---|---|---|---|---|
| 70:30 BP-1: PLA | 25.1 | - | 25.8 | - | 2.48 |
| 80:20 BP-1: PLA | 24.1 | 11.3 | 13.1 | 1.64 | 2.49 |
| 80:18.5:1.5 BP-1: PLA: CP-L01 | 16.6 | - | 14.1 | - | 2.39 |
| 80:20 LHS-1: PLA | 13.8 | 15.1 | 8.8* | 0.76 | 1.24 |
| 85:15 BP-1: PLA | 11.6 | - | 11.6 | - | 1.94 |

*Porosity value was calculated using pycnometer data because displacement density was not available.

The two different methods of void content measurement produced significantly different results. The measurements of porosity using measured densities and mass fractions are higher than that of the CT-scan. These differences can be attributed to variability inherent to the techniques. Most of the porosity values measured via displacement and TGA did not change dramatically when in pellet, ambient print, and TVAC print forms. The CT scan data shows a slight increase in porosity in the TVAC prints compared to ambient. These results could indicate that the porosity was from the extrusion process rather than environmental conditions. Challenges for using CT-scans for porosity measurements include resolution limits and data processing. The size of the volume being scanned, size of the detector, and distance between the sample and detector all affect resolution. There is also inherent noise in the x-ray detection which makes differentiating between similar densities more difficult. These factors can result in smaller porosity being missed by the detectors. Areas with similar densities can make post-processing with reconstruction and porosity detection software challenging. It is likely that the data calculated from

displacement density and mass fraction is a slight overestimate of porosity and CT-scan data is an underestimate. The displacement method can give be skewed by the presence of air bubbles trapped in pores on the surface of the material when it is being measured. This results in and larger volume being displaced in the liquid used for measurement. The actual value for porosity falls within these two measurements.

The advantage of the CT-Scan method for porosity determination is that the data also included a distribution of where the porosity is located. During the printing process in thermal vacuum, the off gassing of material and heat input change the environmental conditions. The initial layer that is printed will be at the coolest temperature and highest vacuum. As the print proceeds, temperature rises and vacuum decreases which could potentially alter the prints properties, especially vacuum induced porosity. Reconstructions made from the CT-Scans showed that there was no significant change in amount or size of porosity towards the top of the build when compared to the bottom. A 3-D reconstruction of an entire sample showing all the pores with a color scale corresponding to pore volume is in Figure 18. A single slice through the 3-D reconstruction showing only pores in that x coordinate is in Figure 19.

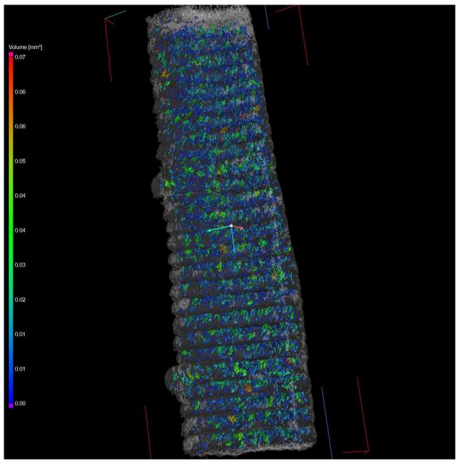


**Figure 18. 3D Reconstruction of 70:30 BP-1: PLA from CT-scan.**

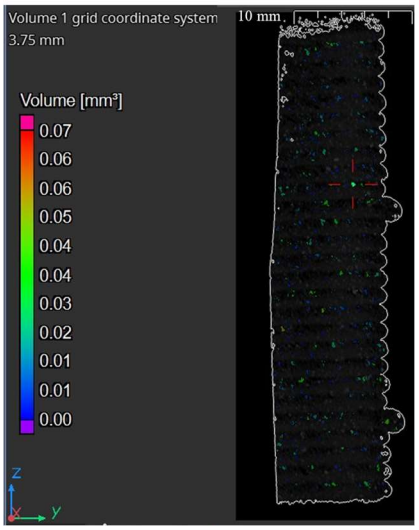



**Figure 19. Slice through 3D reconstruction of 70:30 BP-1: PLA from CT-scan.**

*Surface Morphology Characterization of Beta Materials*

Microscopy of the tensile fracture surfaces of the materials showed porosity, which was consistent with measurements taken using a pycnometer, displacement, CT-scans. On fracture surfaces of samples where the fracture was perpendicular to the layers, porosity was visibly higher at the top of each layer. Figure 20 and Figure 21 shows an example of this with the images oriented with the bottom of the print towards the top of the picture. The perpendicular fracture surfaces were also generally rough indicating a tortuous fracture path. A higher magnification of this transition from porous to more solid is in Figure 22.

In samples with fracture surfaces parallel to the layers, the surface was flat and smoother. When looking at the samples, it was clear that fracture occurred on the interface between layers. An example image is in Figure 23. This is evidence of poor adhesion between each layer of this print.

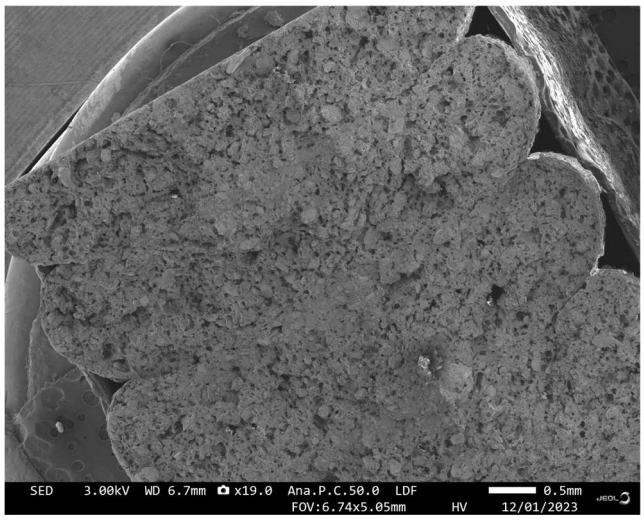



**Figure 20. 85:15 BP-1: PLA tensile fracture surface perpendicular to layers.**

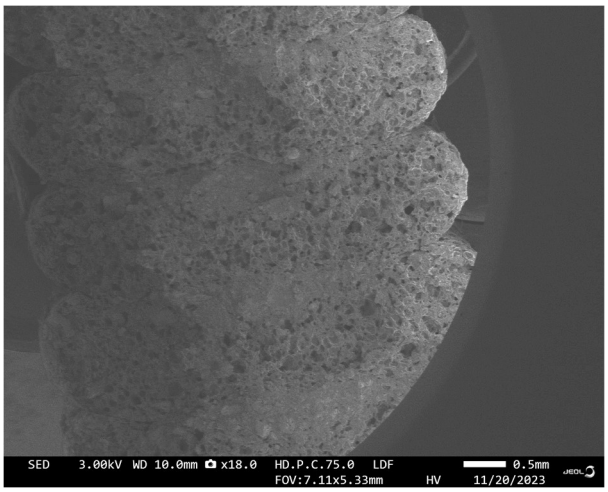



**Figure 21. 70:30 BP-1: PLA tensile fracture surface perpendicular to layers.**

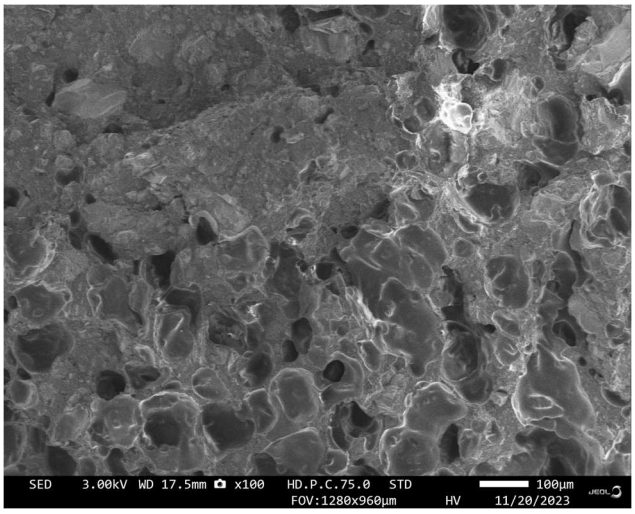



**Figure 22. 70:30 BP-1: PLA higher magnification image of transition in layer from porous to solid.**

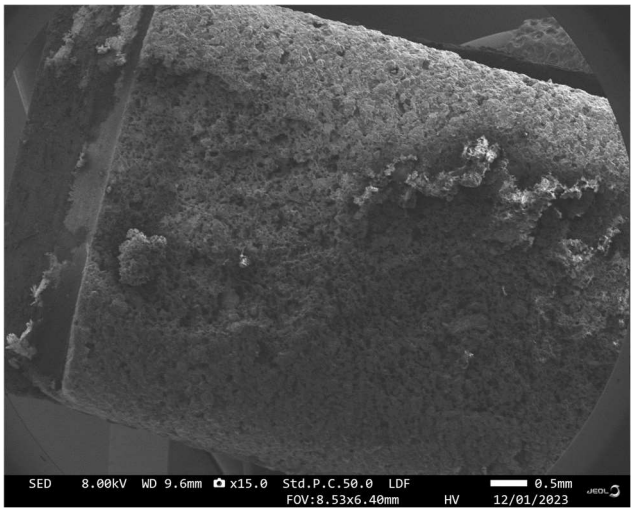



**Figure 23. 80:20 BP-1: PLA tensile fracture surface parallel to layers.**

At higher magnifications, there is evidence of tearing of the polymer matrix and delamination between the regolith and polymer matrix. Smooth regions on the fracture surface are indicative of a lack of fusion between material during printing. This is a result of non-optimal printing conditions for this material. Delamination between the regolith particles and PLA shows that they are not bonding well with each other during printing. Both issues have the potential to be improved with increased melt of the polymer during printing to facilitate bonding and flow. Compatibilizer additives may also improve this condition if available in a form that does not significantly contribute to porosity like the Polypropylene-Graft-Maleic Anhydride did with early polypropylene-based formulations.

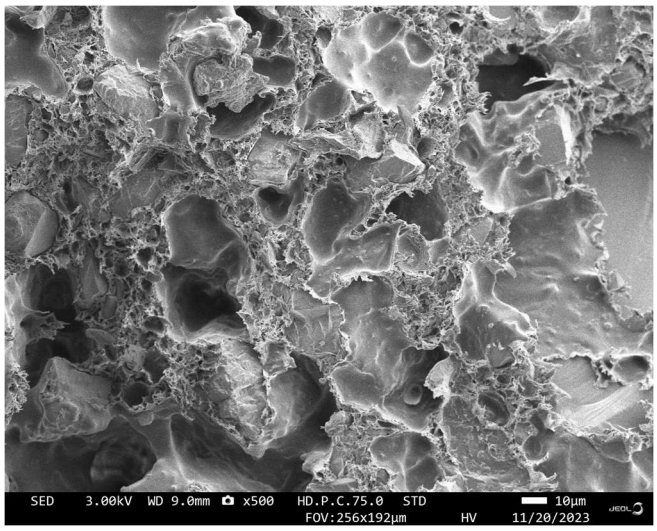



**Figure 24. 85:15 BP-1: PLA higher magnification of fracture.**

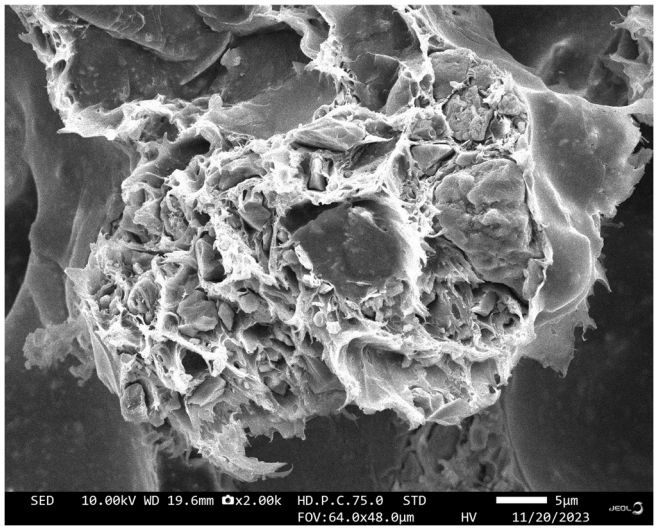



**Figure 25. 85:15 BP-1: PLA ductile tearing of PLA matrix.**

Understanding the relationship between porosity, surface morphology, and mixture ratios of composite materials is crucial for determining their mechanical, thermal, and

structural properties. There is a significant amount of work remaining to understand these relationships.

*Hydrogen Bonding Characterization*

It has been reported that hydrogen bonding interactions in PLA/palygorskite nanocomposites improved the overall tensile properties of the composites due to better dispersion of the palygorskite particles in the PLA matrix [14]. To determine if hydrogen bonding was present, FT-IR spectra of PLA and 80:20 LHS-1: PLA pellets were collected. Unfortunately, the results were inconclusive, and the presence of hydrogen bonding was not readily observed. Follow-on analysis using X-ray photoelectron spectroscopy (XPS) would be beneficial since this is another method for determining hydrogen bonding on surfaces.

*Vacuum Off-Gassing Characterization of Beta Materials*

RGA was used to examine any gases that were present during the TVAC printing process of 85:15 BP-1: PLA and 80:20 LHS-1: PLA cubes. The results are shown in Figure 26 through Figure 29, each plot shows both a representative baseline sample taken before printing and a sample taken during the printing process.

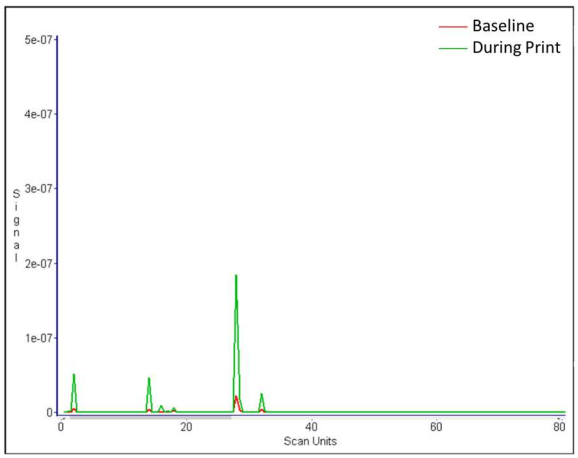



**Figure 26. 85:15 BP-1: PLA major peaks RGA data.**

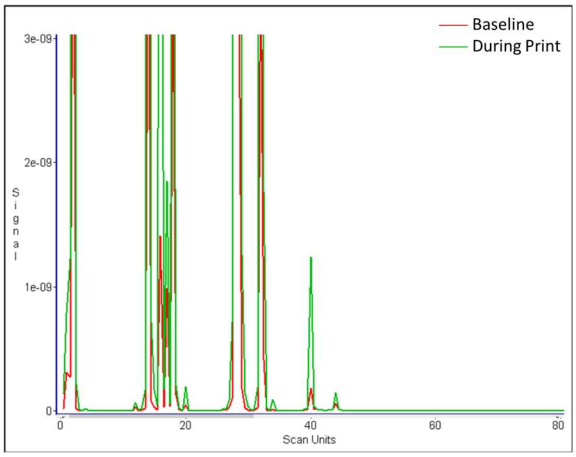



**Figure 27. 85:15 BP-1: PLA minor peaks RGA data.**

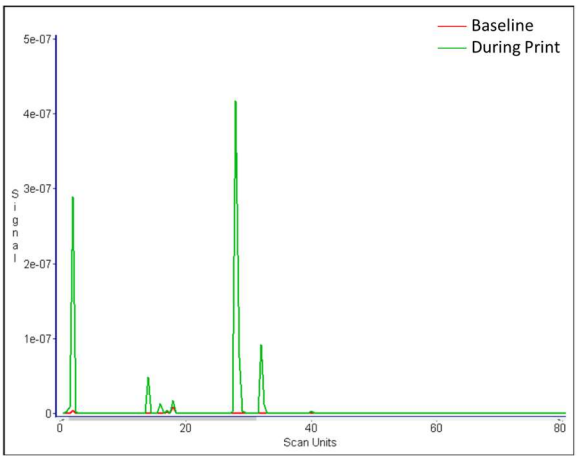



**Figure 28. 80:20 LHS-1: PLA major peaks RGA data.**

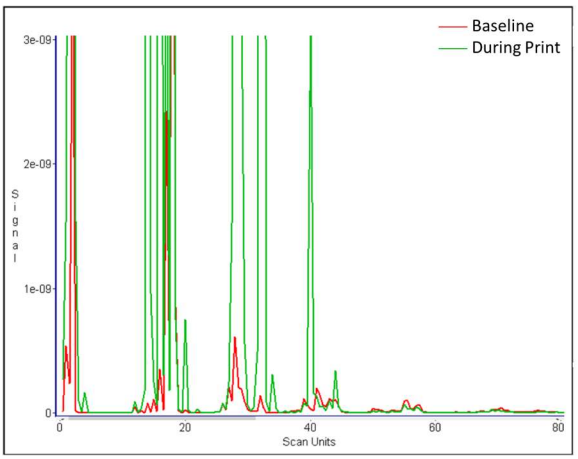



**Figure 29. 80:20 LHS-1: PLA minor peaks RGA data.**

Prominent peaks in the RGA were detected at nitrogen (7, 14, 28 amu), oxygen (8, 16, 32 amu), carbon dioxide (44 amu), and water (18 amu) both before and during the printing process. These are all gas components of air. One known source of the air is from a low-level leak into the vacuum chamber. Another potential source of air is from air trapped in the pellets and released during the printing process. Pellet density is very similar to TVAC printed density, so air release from interior pores is not likely a large contributor. Another potential source of air during the printing process is from air that was condensed on the thermal shroud which may have been liberated when the hot extruder barrel moves within close proximity to the shroud. Water is likely contributed from slow release of moisture adsorbed on interior surfaces of the chamber. Water could also be liberated from residual moisture in the simulants and pellets.

The two formulations have comparable major peaks indicating that the simulants are not likely a major contributor of gases. The figures indicate that there is no substantial off-gassing. Although, one point of interest is the peak at 28 amu (typically nitrogen), which is higher in ratio to the oxygen peak at 32 amu than one would expect to find if it were contributed by air. This may indicate that another compound, perhaps carbon monoxide (CO) is contributing to the increase

at 28 amu.

## 5. Conclusions and Recommendations

NASA's Artemis Program aims to establish a sustained presence on the Moon to provide unprecedented opportunities for scientific exploration and facilitate a thriving lunar economy. Lunar infrastructure will be necessary to provide safe and reliable transportation to/from the surface, blast protection, enhanced mobility, services such as power and communications, and shelter from the environment.

The lunar environment poses serious challenges for construction processes and long-term facilities because of thermal cycling, extreme temperatures, hard vacuum, dust, radiation, moonquakes and meteor impact events. Some of these conditions can be mitigated to a large degree by using regolith as a protective cover over the shelter structure, but construction processes will be exposed to extreme temperatures, vacuum and dust.

The REACT project investigated the use of Regolith Polymer Composites (RPC) as a building material in a Fused Filament Fabrication (FFF) additive construction process under simulated lunar DTVAC conditions. Basic feasibility of the construction system and materials was proven with PLA based formulations. PP experienced significant porosity when extruded in vacuum. The source of the porosity may be from residual moisture in the regolith simulant. It is not likely due to gas products generated from molten polymers because the RGA data did not show these compounds. Porosity data was highly variable depending on the measurement method and ranged from 25.8 – 8.8 % for PLA formulations (excluding CT scan measurements). Porosity was concentrated at layer interfaces suggesting that improvements in the deposition process could reduce the effect. Notably, the porosity values did not change dramatically when in pellet, ambient print, and TVAC print forms indicating that the porosity was likely from the extrusion process rather than environmental conditions.

RPC materials extruded in DTVAC can yield a solidified form with strength properties that are sufficient for structural applications. Flexural modulus and strength controlled LINA's structural design. The 80:20 LHS-1: PLA wt.% formulation was the Beta material of choice because of the simulant being more representative of the south polar regions and good strength properties. The 80:20 LHS-1: PLA wt.% formulation had a flexural modulus of 5.3 GPa and flexural strength of 24 MPa in the 0° orientation. For comparison, readily available lumber used in typical wood construction generally has a modulus of 6-10 GPa and a flexural strength of 4-8 MPa. Typical flexural strength of unreinforced concrete is similar and are in the range of 3-5 GPa. These findings highlight the potential suitability for lunar construction. However, the material properties reported here result from testing in ambient temperatures. An analysis of the expected temperatures of the materials on the Moon is necessary along with strength testing across that thermal range.

Future development should use higher fidelity simulants with representative chemical makeup, particle size distribution, and morphology with agglutinates. These factors can affect feeding rates, energy consumption, wear, and material strength properties. Lunar regolith will also contain nanophase iron and solar wind deposited elements which may affect the resultant material.

The REACT project was only able to assess PP and PLA as binders for RPC because of cost and schedule constraints. PEI showed promise because of its superior properties and space flight heritage, but the project's extruder could not handle the elevated processing temperature. PEI should be a priority for testing in the future and the next generation extruder should be designed for its use. In general, data on various polymer properties at cryogenic temperatures were scarce in the literature. A more exhaustive assessment of available polymers should be completed. Special attention should be paid to polymers that have the potential to be produced from lunar in-situ materials. An effort to design polymers specifically for the lunar environment could significantly improve RPC properties.

Further testing is needed to assess RPC's performance over the expected lifecycle of a lunar facility. Exposure to the Moon's challenging environment may limit the useful life of RPC because of issues like thermal cycling fatigue, mass loss due to outgassing, and UV degradation.

The properties of additively constructed RPC are not solely dependent on the intrinsic material properties but are also significantly influenced by various factors involved in the printing process. Parameters such as processing temperatures, print speeds, and extrusion rates play pivotal roles, and have the potential to induced defects. Additional RPC materials development and construction system testing should use systems that are more representative of a flight demonstration mission. Furthermore, the two-step material compounding then printing process should be reduced into a single step where compounding occurs "on the fly" during construction. This approach has the added benefit of eliminating water usage from the process.

Though development work remains towards flight like extrusion systems, polymer selection/characterization, and lifecycle considerations; the data provides a useful benchmark for the first high-fidelity ISRU based additively constructed structure in thermal vacuum conditions. Additive construction using RPC shows a lot of promise because of its proven capability to create complex geometry in simulated lunar environments, adequate strength properties, and demonstration at scale during the NASA 3D printed habitat centennial challenge [15]. A more comprehensive and systematic testing regimen is requisite to establish a robust correlation between the material's strength properties and its thermal properties, with an analysis

comparing the print process parameter effects on the resultant material using flight-like hardware. This in-depth approach would enable the development of an optimized, repeatable solution that accurately reflects the complex interplay between material, process, and performance in 3D printed samples. The progress made during the REACT project warrants further development towards a small-scale flight demonstration mission.

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## APPENDIX – SELECTED POLYMER PROPERTIES

### Table 8 – Selected Polymer Properties

| Material | Thermal | | | | Mechanical | | | | | | Other | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | 77K | | | RT | | | Cost | Outgassing | | Ref. |
| | CTE | HDT | $T_m$ | $T_g$ | $\varepsilon_f$ | E | $\sigma_{UT}$ | $\varepsilon_f$ | E | $\sigma_{UT}$ | | TML | REF | |
| | um/m°C | °C | °C | °C | % | GPa | MPa | % | GPa | MPa | USD/ kg | % | # | # |
| PP (H110MA) | 95 | 104 | 235 | -10 | 3.6 | 1.65 | 45 | 100 | 0.60 | 30 | 1.23 | 0.34 | 16916 | 15, 21, 25, 27 |
| PLA | 114 | 56 | 156 | 57 | 5.7 | 1.67 | 87 | 6.4 | 1.17 | 41.6 | 2.99 | 0.56 | 35082 | 14, 32 |
| PEI (Ultem | 65 | 169 | 340 | 177 | 1.4 | 7.48 | 101 | 70.0 | 2.15 | 69 | 18.90 | 0.4 | 32863 | 9, 26, 28 |
| FEP (Teflon | 120 | 140 | 259 | 80 | 1.9 | 6.80 | 131 | 32.5 | 0.60 | 27 | 11.25 | 0.01 | 32098 | 3, 17 |
| PEEK | 65 | 186 | 343 | 143 | 8.0 | 5.40 | 200 | 70.0 | 3.50 | 95 | 66.10 | 0.3 | 14250 | 7, 29 |
| PI (Aurum PL450C) | 55 | 238 | 388 | -273 | 16.0 | 1.59 | 184. 5 | 90.0 | 2.76 | 118 | 27.50 | 0.59 | | 13, 23, 24 |
| PETG-CF | 29 | 77 | 229 | 80 | 1.2 | 7.06 | 82.8 | 3.0 | 4.93 | 55.5 | 64.00 | 0.35 | 18909 | 1, 20 |
| PBT | 140 | 165 | 224 | 32.5 | 5.0 | 5.8 | 292 | 170 | 2.30 | 55 | 1.90 | 0.24 | 22801 | 5, 18, 22 |
| PHA/PHB | 210 | 92 | 176.5 | 13.5 | | | | 15.5 | 3.75 | 37.5 | 4.15 | | | 12 |
| Epoxy Resin | 99 | 192 | - | 117 | 2.6 | 7.23 | 185 | 4.5 | 2.41 | 67.3 | 3.57 | 0.49 | 4392 | 2, 16 |
| Polyamide | 146.5 | 183 | 215 | 50 | 1.2 | 2.27 | 28.8 | 4.1 | 1.83 | 55.7 | 2.29 | 1.50 | 927 | 4, 19 |
| HDPE | 152 | 85 | 133.5 | -108 | 5.7 | 7.05 | 160 | 1205 | 1.08 | 28.5 | 1.16 | 0.23 | 33547 | 8, 16 |
| LDPE | 288 | 42 | 106.5 | -108 | 2.0 | | 118 | 445 | 0.23 | 10.2 | 1.28 | 0.21 | 19489 | 10, 30 |
| UHMW | 297 | 75 | 131.5 | -108 | 6.0 | | 160 | 438 | 0.93 | 43.5 | 2.28 | 0.22 | 6470 | 11, 30 |
| XLPE | 180 | 80.5 | 139.5 | -108 | 2.4 | | 135 | 390 | 1.20 | 15.8 | 1.48 | | | 6, 30 |

| Source Key: |
|---|
| Published test data |
| Manufacturer data (TDS) |
| Online database (ANSYS/MAPTIS) |
| NASA Outgassing Data [31] |
| Interpolated value |

**Notes:**

- Where range of values is presented, an average value is given; properties may vary based on polymer grade and test conditions
- Interpolated values assume linear elastic to failure behavior, may not be valid for all polymers
- 3D printed materials (PLA, PEI, PETG-CF) may exhibit anisotropic properties, data may not be 100% comparable to molded polymers
- When possible, effort was made to match grades from published academic sources with manufacturer data
- Heat distortion temperature (HDT) taken at 0.45 MPa

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